

Cross-Impact of Order Flow Imbalance in Equity Markets

Assignment Answers

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1. Motivation for Multi-Level OFI Measurement

The order *flow imbalance* (OFI) measures the net excess of buying vs. selling orders in the limit order book (LOB). Traditionally, OFI has been measured using only the best (top) level of the order book, but the authors extend this concept to multiple depth levels. The motivation for incorporating multiple levels is that deeper layers of the LOB also contribute to price formation and contain valuable information about supply and demand beyond the best bid and ask. By measuring OFI at each level $\ell = 1, 2, \dots, L$, one captures a more complete picture of the net order flow. For instance, if large volumes rest just behind the best quotes, they can absorb incoming orders and dampen immediate price moves; conversely, if deeper levels are thin, even moderate order flow can move the price. Thus, a multi-level OFI provides a more informative signal of buying/selling pressure than a single-level OFI.

Mathematically, let $\Delta V_{\text{bid}}^{(1)}$ and $\Delta V_{\text{ask}}^{(1)}$ denote the net changes in volume available at the best bid and best ask, respectively, over a short interval. The **best-level OFI** can be defined as the signed volume imbalance at the top of the book:

$$\text{OFI}^{(1)} = \Delta V_{\text{bid}}^{(1)} - \Delta V_{\text{ask}}^{(1)}, \quad (1)$$

which is positive when buy orders dominate (either new buys added or sell orders lifted from the ask) and negative when sell orders dominate. A **deeper-level OFI** at level ℓ is defined analogously as $\text{OFI}^{(\ell)} = \Delta V_{\text{bid}}^{(\ell)} - \Delta V_{\text{ask}}^{(\ell)}$, capturing the net order flow at the ℓ -th level of the LOB. The authors measure OFI across the top L levels, forming a vector $\mathbf{OFI}(t) = (\text{OFI}^{(1)}(t), \text{OFI}^{(2)}(t), \dots, \text{OFI}^{(L)}(t))$ of multi-level imbalances. Because average depth and order arrival rates can differ by level, each level's OFI may be scaled by the typical depth at that level to normalize the impact.

An important finding is that these multi-level OFIs are highly correlated with each other (often exceeding 75% correlation between levels). This indicates that order flow pressure tends to be

systemic across the book: large incoming buy (sell) flows often impact several top levels in the same direction. In fact, the first principal component of the **OFI** vector explains about 89% of the variance across levels. This motivates constructing an **integrated OFI** that aggregates information from all L levels into a single variable. Using principal components analysis (PCA), the authors define the integrated OFI as the first principal component of the multi-level OFI vector. If w_ℓ is the weight for level ℓ from the first principal component (normalized so that $\sum_{\ell=1}^L w_\ell = 1$), the integrated OFI is given by

$$\text{OFI}^{(\text{int})}(t) = \sum_{\ell=1}^L w_\ell \text{OFI}^{(\ell)}(t), \quad (2)$$

which combines multi-level imbalances into one aggregate measure. The benefit of this approach is evidenced by a significantly improved ability to explain price movements. Empirically, using an OFI that incorporates multiple depths yields a higher R^2 in explaining short-term price returns (e.g. about 87% with integrated multi-level OFI, versus 71% using only best-level OFI). In other words, much of the price impact that was not captured by the best-level order flow is accounted for when including deeper levels. Indeed, once information from multiple levels is integrated into the OFI, adding cross-asset terms yields little additional explanatory power for same-time price changes. This underscores the motivation: measuring OFI at multiple depths provides a more complete and predictive summary of order flow pressure on prices.

2. LASSO vs. OLS for Cross-Impact Estimation

When extending the analysis to *cross-impact* — how one asset's order flow affects another asset's returns — the authors face a high-dimensional regression problem. For a universe of N stocks, a linear cross-impact model for stock i 's return $r_i(t)$ at time t may be written as:

$$r_i(t) = \alpha_i + \sum_{j=1}^N \beta_{ij} \text{OFI}_j(t) + \varepsilon_i(t), \quad (3)$$

where $\text{OFI}_j(t)$ is the order flow imbalance of stock j in the same interval. The coefficient β_{ii} represents the *self-impact* of stock i 's own order flow on its return, while β_{ij} ($j \neq i$) are *cross-impact* coefficients capturing spillover effects from other stocks' order flow. In a fully specified model, stock i could have up to $N - 1$ cross-impact predictors in addition to its own OFI. For large N or short estimation windows, an ordinary least squares (OLS) regression becomes infeasible and unstable: the number of parameters can exceed the number of observations, violating basic OLS requirements. Moreover, the different stocks' OFI streams are often highly correlated (e.g. many stocks might be influenced by common market-wide order flow surges), leading to severe multicollinearity. Including many correlated predictors with OLS can result in overfitting and statistically insignificant estimates for the cross-impact coefficients.

To address these challenges, the authors employ the **Least Absolute Shrinkage and Selection Operator (LASSO)** for estimating cross-impact models. LASSO is a regularization technique

that adds an L^1 penalty on regression coefficients, effectively performing variable selection by driving many coefficients to zero. The regression is formulated as minimizing a penalized sum of squared errors:

$$\min_{\{\beta_{ij}\}} \sum_{t=1}^T \left(r_i(t) - \alpha_i - \sum_{j=1}^N \beta_{ij} \text{OFI}_j(t) \right)^2 + \lambda \sum_{j=1}^N |\beta_{ij}|, \quad (4)$$

where $\lambda > 0$ is a tuning parameter that controls the strength of regularization. The L^1 penalty $\sum_j |\beta_{ij}|$ encourages sparsity in the solution, meaning only a relatively small subset of other assets will be identified as having a significant cross-impact on stock i . This aligns with the authors' assumption that each stock is mainly influenced by a few other stocks rather than by the entire market. By using cross-validation to choose λ , the model avoids overfitting while retaining the most predictive OFI features for each stock's returns. In contrast, a plain OLS regression including all N order flow variables would likely overfit or fail to produce stable estimates in this high-dimensional setting. The LASSO approach yields a more interpretable and parsimonious model of cross-impact, as it zeroes out negligible cross-effects and highlights the truly influential cross-asset OFI predictors. Thus, the authors favor LASSO over OLS to reliably estimate cross-impact coefficients under the constraints of high dimensionality and multicollinearity.

3. OFI vs. Trade Volume as Predictors of Short-Term Returns

Order flow imbalance (OFI) is regarded as a superior predictor of short-term price returns compared to raw trade volume because OFI captures the *directional* pressure of trading, not just the quantity. Trade volume (e.g. number of shares traded in a minute) is an *unsigned* measure — it aggregates buys and sells without distinction. In contrast, OFI is a *signed* measure that netted buy orders against sell orders. To illustrate, let V^{buy} be the total volume of buyer-initiated trades (market buys) in a short interval, and V^{sell} the volume of seller-initiated trades (market sells). The total trade volume is

$$V_{\text{total}} = V^{buy} + V^{sell}, \quad (5)$$

whereas the order flow imbalance (from trades) would be

$$\text{OFI}_{\text{trades}} = V^{buy} - V^{sell}. \quad (6)$$

Price impact theory and empirical evidence indicate that price changes are primarily driven by the imbalance (the difference between buying and selling pressure), rather than the sheer amount of volume traded. If an equal volume of shares is bought and sold, $V^{buy} = V^{sell}$, the net imbalance is zero and one would expect little price movement despite high turnover. Conversely, if buying significantly exceeds selling ($\text{OFI}_{\text{trades}}$ is strongly positive), the price is likely to rise in the short term; similarly, excess selling (negative imbalance) predicts downward pressure. Thus, the imbalance $V^{buy} - V^{sell}$ is more predictive of the next price change than $V^{buy} + V^{sell}$.

Moreover, the OFI metric used by the authors goes beyond just executed trade volumes. It also incorporates limit order placements and cancellations at the best quotes (and even deeper levels, as discussed in Section 1). This means OFI accounts for liquidity supply and demand that may not immediately result in trades but still affect the price (for example, a large buy limit order added on the bid side increases buying pressure even if not instantly filled). Empirical studies have confirmed that short-horizon price returns correlate much more with such order flow imbalances than with raw volume. Cont *et al.* (2014) found that over short intervals, price changes are mainly driven by OFI. In other words, once one knows the net order imbalance, the total volume offers little additional information about the price move. Additionally, subsequent research has shown that models using OFI as the predictor outperform those using traditional volume or even the full order book state in forecasting short-term returns. For instance, deep learning models trained on sequences of OFI features achieved better predictive accuracy than models fed with complete LOB data or past returns, highlighting that the essential information for short-term price prediction is contained in the order flow imbalance.

In summary, OFI is a better predictor of short-term returns than trade volume because it encapsulates the **direction and pressure** of trades (and order book changes) driving price movements. Trade volume alone fails to distinguish whether trading activity was predominantly buyer-initiated or seller-initiated, whereas OFI quantifies this imbalance. The authors therefore focus on OFI-based predictors, leveraging the fact that imbalance-based models can explain and forecast price changes far more effectively than volume-based metrics. This choice is supported by both theoretical frameworks (e.g. Kyle's model of price impact, where price responds to net order flow) and empirical results demonstrating the higher predictive power of OFI for short-term price dynamics.